JON KENTON TARSUS BRODZIAK

NOAA Fisheries, Pacific Islands Fisheries Science Center
1845 Wasp Boulevard, Building 176
Honolulu, Hawaii 96818 USA

Email: Jon.Brodziak@NOAA.GOV
Telephone: (808) 725-5617
Linkedin: https://www.linkedin.com/in/jon-brodziak-420a2535/
Google: https://scholar.google.com/citations?user=nYE0AWAAAAAJ&hl=en
Researchgate: https://www.researchgate.net/profile/Jon-Brodziak

June 2022

EDUCATION

University of California, Davis

- Ph.D. Applied Mathematics 1990
- M.Sc. Applied Mathematics 1987
  Earl Anthony Fellowship Award 1985-1986

University of Pennsylvania

- B.A. Mathematics 1985
  Magna cum laude
  Philosophy Minor
  Mathematics Honor Society President 1984-1985

CURRENT RESEARCH

- Population Dynamics, Stock Assessment, Ecosystem-Based Fisheries Management, Mathematical Ecology, Bayesian Statistics, Simulation Modeling, Numerical Methods

PROFESSIONAL EXPERIENCE

SENIOR STOCK ASSESSMENT SCIENTIST

Mathematical Statistician, ZP IV, Band 3 1-May-2006 to present
National Marine Fisheries Service, Pacific Islands Fisheries Science Center

- Primary duties are:
  - Serve as senior stock assessment scientist
  - Conduct stock assessments and lead stock assessment teams
  - Investigate factors that influence the abundance and dynamics of marine populations
  - Improve modeling of marine ecosystem dynamics
  - Develop frequentist and Bayesian stock assessment models and software
  - Evaluate the effects of population, community, and environmental factors on the productivity of insular and highly migratory fishery resources
  - Convene international scientific review panels
  - Serve as chair or USA delegation lead for domestic and international working groups
  - Disseminate scientific information to stakeholders including the public, the fishing industry, the Western Pacific Fishery Management Council, the Western and Central Pacific Fisheries Commission, the Inter American Tropical Tuna Commission and the International Council for the Exploration of the Sea.
- PIFSC Member of NMFS National Assessment Methods Working Group, 2006-present.
- Chair of the NMFS National Assessment Methods Working Group, 2015-2017.
- Developer of AGEPRO, SRFIT, YPR, STATCAM and MAS software modules in the NOAA Fisheries Toolbox, 2006-present. Available at: http://nft.nefsc.noaa.gov/.
- Member of NMFS National Standard 1 Guidelines Technical Working Group, 2017-present.
- Member of NMFS National Fisheries Integrated Modeling System Working Group, 2020-present: https://github.com/NOAA-FIMS.
- Project Leader for NMFS Metapopulation Assessment System Working Group and r4MAS R package, 2015-present: https://github.com/nmfs-fish-tools/r4MAS.
- Acting PIFSC Member of NMFS Management Strategy Evaluation Working Group, 2017-2021.
- Member of the NMFS Stock Assessment Improvement Plan Working Group, 2013-2018.
- Affiliate Faculty Professor, Marine Biology Graduate Program, University of Hawaii, 2013-present.
- Ph.D. Committee Member, Zack Oyafuso, UH Marine Biology Graduate Program, 2015-present.
- Ph.D. Committee Member, Haruko Koike, UH Marine Biology Graduate Program, 2012-2017
- M.Sc. Committee Member, Cassandra Pardee, Hawaii Pacific University, 2013-2014.
- NMFS Sea Grant Fellow Mentor, Quang Hyuhn, VIMS, 2016-2017.
- NMFS Sea Grant Fellow Mentor, Zack Oyafuso, UH Manoa, 2015-2016.
- Acting PIFSC Stock Assessment Program Leader, August-October, 2015.
- Interim Chair of the Statistics Working Group of the International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific [ISC], 2020-2021.
- Chair of the Billfish Working Group of the International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific, 2011 -2019.
- Member of USA delegation at the International Scientific Committee on Tuna and Tuna-Like Species in the North Pacific [ISC] Plenary Meeting, 2007-present.
- Member of the Joint EU, NOAA, IMR NewSciFish Working Group, 2018-present.
- Member of the International Council for the Exploration of the Sea [ICES] Working Group on Recruitment Forecasting in a Variable Environment, 2014-2017.
- Convener of the Stock Assessment Theme Session at the 7th to the 14th Regular Meetings of the Scientific Committee of the Western and Central Pacific Fisheries Commission, 2011-2018.
- Convener of Symposium "*Population Dynamics and Sustainable Fisheries for Highly-Migratory Large Pelagic Species*" at the American Fisheries Society 2014 Annual Meeting, Quebec City.

- Invited participant in NMFS Data Limited Methods Workshop, January 2014.
- Invited participant in NMFS National Standard 1 Guidelines Workshop, December 2012.
- Invited speaker at National Academies Ocean Studies Board Committee on *Evaluating the Effectiveness of Stock Rebuilding Plans of the 2006 Fishery Conservation and Management Reauthorization Act*, Seattle, July 2012.
- Invited speaker at NOAA-Shanghai Ocean University-ISC Science Symposium at Chinese Academy of Fisheries Science, Shanghai, April 2012.
- Member of NOAA Fisheries Climate Change Vulnerability Working Group, 2011- 2013.
- Invited participant in the Pacific Integrated Ocean Observing System Biological Observations Project, May 2011.
- Invited participant in NMFS Annual Catch Limit Science Workshop, February 2011.
- Head of USA Delegation at the Joint Tuna RFMOs Meeting of Experts to Share Best Practices on the Provision of Scientific Advice, Barcelona, Spain, June 2010.
- Invited lecturer at Center for Sustainability Science, University of Hokkaido, Hakodate, April 2010.
- Invited speaker SEDAR "Uncertainty Procedures Workshop", Charlotte, February 2010.
- Participant in NOAA Leadership Training Program, September 2009.
- Co-convener of NMFS Assessment Methods Working Group workshop "Estimating natural mortality in stock assessment applications", Seattle, August 2009.
- Participant in NMFS 2nd National Ecosystem Modeling Workshop, Annapolis, August 2009.
- Invited lecturer at Shanghai Ocean University, Shanghai, April 2009.
- Convenor of Symposium "Sustainable fisheries for highly migratory large pelagics" at American Fisheries Society 2008 Annual Meeting.
- Invited speaker at 5th World Fisheries Congress Stock Assessment Methods Session, October 2008.
- Invited lecturer at Tokyo University, Hokkaido, June 2008.
- Invited participant in Stock Assessment Program Review Panel for the Southwest Fisheries Science Center, April 2008.
- Member of NOAA MSA-406 Steering Committee and contributor to NOAA Tech. Mem. 96 "The State of Science to Support an Ecosystem Approach to Regional Fishery Management".
- Invited participant in Hawaiian Archipelago Marine Ecosystem Research Panel, August 2007.
- Invited participant in Pelagic Longline Catch Rate Standardization Workshop. Honolulu, 2007.
- Invited participant in NOAA Satellite Data Course. Corvallis and NOAA-NASA Integrating Satellite Data Products into Ecosystem-Based Management of Living Marine Resources Workshop. Pacific Grove, May-August 2006.
- Participant in NMFS National Stock Assessment Workshops 5 through 13.

LEAD STOCK ASSESSMENT SCIENTIST

Operations Research Analyst, ZP IV 1-Feb-1999 to 30-Apr-2006
National Marine Fisheries Service, Northeast Fisheries Science Center

- Primary duties were:
  - Serve as senior stock assessment scientist on the New England Fishery Management Council's Groundfish Plan Development Team
  - Serve as lead assessment scientist for northeast USA haddock, silver hake, red hake, and Atlantic halibut resources
  - Apply and develop frequentist and Bayesian stock assessment models and software.
  - Conduct research to evaluate the effects of population, community, and environmental factors on the productivity of commercial fishery resources in the northwest Atlantic
  - Direct technical staff to assist with assessment related research
  - Participate in scientific research surveys

- Disseminate scientific information to the public, industry, and the New England and Mid-Atlantic Fishery Management Councils

- Developer of AGEPRO, SRFIT, and STATCAM software modules in NOAA Fisheries Toolbox during 2000-2006. Available at: http://nft.nefsc.noaa.gov/
- Lead stock assessment scientist on New England Fishery Management Council's [NEFMC] Groundfish Plan Development Team, 2001-2006, responsible for biological analyses of Amendment 13 and Frameworks 36 through 43 of the Northeast Multispecies Fishery Management Plan of the NEFMC.
- Co-Chair of NEFSC Ecosystem Status Working Group (2001-2004) and co-editor of *Status of the Northeast U.S. Continental Shelf Ecosystem. NEFSC Ref. Doc. 02-11.* Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0211/
- Invited participant in ASMFC Horseshoe Crab Stock Assessment Workshop, Philadelphia, January 2006.
- Adjunct Professor, Wildlife and Fisheries Conservation Program, University of Massachusetts Amherst, 2002-2006.
- M.Sc. Committee member, Douglas Sigourney, Wildlife and Fisheries Conservation Program, University of Massachusetts Amherst, 2003-2005.
- Invited participant in NOAA MPA Benthic-Pelagic Linkages Workshop, Monterey, November 2005.
- Invited participant in NMFS Workshop on Ecosystem-Based Decision Support Tools for Fisheries Management, Key Largo, February 2005.
- Instructor for the course Bayesian Analysis of Catch Rates, URI Sea Grant College Program, 2004. Available at: http://seagrant.gso.uri.edu/reg_fish/gear/workshops.html
- Lead haddock biologist on USA/Canada Transboundary Resource Assessment Committee, 2001-2006.
- Member of Stellwagen Bank Marine Sanctuary Ecosystem Management Working Group, 2003-2004.
- Methods developer and software engineer for age-structured reference point estimation and projection models used by the Working Group on Re-evaluation of Biological Reference Points for NE groundfish, 2002. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0204/
- Chair of the Stock Assessment Review Panel for west coast sablefish, July 2001. Available at: http://nft.nefsc.noaa.gov/
- Stock assessment scientist for haddock, halibut, and silver hake resources, 1999-2006.
- Stock assessment model developer for New England redfish, winter flounder, and witch flounder resources, 2000-2005.
- Invited speaker at the 4th Mote International Symposium in Fisheries Ecology: *Confronting Trade-offs in the Ecosystem Approach to Fisheries Management*, November 2002.
- Member of NEFMC's Silver Hake Plan Development Team and Monitoring Committee, 1999-2002.
- Field scientist on 9 research cruises and 3 commercial longline trips for a total of 80 days at sea.
- Participant in 12 Northeast Regional, SEDAR, and STAR Groundfish Stock Review Assessment Workshops during 1999-2006.
- Presenter at 6th, 7th, and 8th NMFS National Stock Assessment Workshops, 2001-2004.
- M.Sc. Thesis Committee member. Jennifer McCarthy, Department of Fisheries and Wildlife of University of Massachusetts, Dartmouth, 2002-2004.
- Member of Squid Research and Scientific Committee of the State of California, USA, August 1999-2001.
- Invited speaker at the 3rd Mote International Symposium in Fisheries Ecology: *Targets, Thresholds, and the Burden of Proof in Fisheries Management*, November 2000.
- Review panel member for the Canadian Regional Assessment Process Meeting, Maritimes Provinces, Halifax, Nova Scotia, November 1999.

STOCK ASSESSMENT SCIENTIST

Mathematical Statistician, ZP III 1-Sep-1995 to 31-Jan-1999
National Marine Fisheries Service, Northwest Fisheries Science Center

- Primary duties were:
  - Assess the Dover sole resource.
  - Develop abundance indices for deep-water groundfish based on fishery catch rates from commercial logbook data
  - Develop stock assessment methods and software.
  - Participate in scientific surveys, stakeholder meetings, and scientific workshops
  - Disseminate information to the public, industry, and the Pacific Fishery Management Council
- Chair of the Stock Assessment Review Panel for Blackgill and Chilipepper Rockfish, June 1998.
- Invited speaker at the Industry and Science Conference: *Working Together for West Coast Groundfish*, July 1998.
- NWFSC member of the Groundfish Management Team to the PFMC, 1997-1999.
- Adjunct Professor, Department of Fisheries and Wildlife, Oregon State University, 1996-1999.
- M.Sc.Committee member, John Harms, Department of Fisheries and Wildlife, Oregon State University, 1997-1999.
- Co-Developer of stock assessment review guidelines for PFMC groundfish stock assessments (with Larry Jacobson), 1996-1999.
- Member of assessment working groups for sablefish and shortspine thornyhead, 1998.
- Instructor for *Fishery Stock Assessment Methods* in the Department of Fisheries and Wildlife, Oregon State University, 1998.
- Invited speaker at the Cephalopod International Advisory Council Workshop on Cephalopod Ecology and Fishery Management, Cape Town, South Africa, 1997.
- Invited speaker and panel member at California Cooperative Fisheries Investigation Symposium on Market Squid, October 1997.
- Lead stock assessment scientist for Dover sole and liaison for the PFMC Groundfish Management Team.
- Participant in the West Coast Groundfish Stock Assessment Planning and Review Meetings, 1996-1998.
- Field scientist participating in 2 research cruises for a total of 9 days at sea.

STOCK ASSESSMENT SCIENTIST

Operations Research Analyst, ZP III 15-Oct-1990 to 31-Aug-1995
National Marine Fisheries Service, Northeast Fisheries Science Center

- Primary duties were:
  - Assess the status of Atlantic mackerel, long-finned squid, Atlantic butterfish, and short-finned squid resources in the Northwest Atlantic
  - Participate in scientific working groups, meetings, and workshops and the dissemination of information to the public, industry, and Mid-Atlantic Fishery Management Council
- Stock assessment scientist for the Atlantic mackerel, long-finned squid, Atlantic butterfish, and short-finned squid resources for the MAFMC.
- Participant in a total of 11 Northeast Regional Stock Assessment Workshops, 1990-1995.
- Member of scientific working groups (WG): International Council for the Exploration of the Sea [ICES] Shellfish Committee; ICES Cephalopod Fisheries and Life History WG; NMFS Ad Hoc Risk Assessment WG; the NEFSC Sea Sampling, Assessment Methods, Invertebrate, Pelagic, and Coastal WGs.
- Technical analyst for Amendment 7 of the Northeast Multispecies Groundfish Fishery Management Plan for NEFMC and developer of risk analyses for groundfish yields under Amendment 7

management options.
- Technical analyst for the development of multispecies models and bioeconomic analyses of Groundfish Plan Amendment 5.
- Invited participant in International workshops on the status of harbor porpoises, 1992 and 1994.
- Participant in the 2nd, 3rd , and 4th NMFS National Stock Assessment Workshops, 1992-1994.
- Participant in ICES Multi-species WG Meeting, December, 1990.
- Field scientist participating in 6 NEFSC research surveys during 1991-1995 for 62 days at sea.

TEACHING EXPERIENCE

University of Hawaii, Manoa

- Instructor. Marine Biology 691, Graduate Research Seminar with Marine Biology Faculty. 2015-2016.
- Lecturer. Marine Biology 602, Processes and Impact, Marine Biology Graduate Program. 2013-2014.
- Lecturer. Zoology 739, Ecosystem Approaches to Fisheries, Marine Biology Graduate Program. 2013.

University of Rhode Island

- Instructor. Dynamics of Exploited Marine Populations. 2004-2005.
- Lecturer. URI Fishery Industry Catch Comparison Workshop. 2004.

University of Massachusetts

- Instructor. Dynamics of Exploited Fish Populations. School of Marine Science and Technology. 2003.

Jackson State University

- Instructor. NMFS Population Dynamics and Stock Assessment Summer Classes. Jackson State Department of Biology. 2002-2004.

Oregon State University

- Instructor. OSU 499/599. Fishery Stock Assessment Methods. 1998.

University of California, Davis

- Instructor. Differential and Integral Calculus. UC Davis Department of Mathematics. 1987-1990.
- Teaching Assistant. UC Davis Department of Mathematics. 1985-1990.

University of Pennsylvania

- Mathematics Tutor. Pi Mu Epsilon Undergraduate Mathematics Honor Society. 1984-1985.

PROFESSIONAL HONORS AND AWARDS

NOAA Bronze Medal Award. 2020. For enhancing stock assessment methodologies with the unprecedented inclusion of research video-camera surveys and fishing industry engagement.

NOAA General Counsel's Award. 2004. For exceptional performance and significant contributions to the Office of the General Counsel.

NOAA Bronze Medal Award. 2004. For the successful and timely completion of Amendment 13 to the Northeast Fishery Management Plan.

NOAA Administrator's Award. 2004. For outstanding leadership in the successful development of Amendment 13 to the Northeast Multispecies Fishery Management Plan.

NOAA Bronze Medal Award. 2003. For scientific advances in restoring New England groundfish stocks via population dynamics research and management advisory services.

NEFSC Unit Award. 1997. In Recognition of Outstanding Achievements In Support of Amendment 7 to the Northeast Multispecies Fishery Management Plan.

NOAA Group Citation. 1994. For Outstanding Stock Assessment Research.

NEFSC Outstanding Performance Award. 1993 and 1995. Northeast Fisheries Science Center Director.

Fellow of the American Fisheries Society 25 Year Membership Award. 2016.

Fellow of the American Institute of Fishery Research Biologists Award. 2003.

PUBLICATIONS

Methot, R., **Brodziak, J**., et al. **In review**. Overview of Approaches and Updated Guidance for Estimating Status Determination Reference Points and their Proxies in Accordance with the National Standard 1 Guidelines. NOAA Fisheries Technical Memorandum.

Kaplan, I., Gaichas, S., Stawitz, C., Lynch, P., Marshall, K., Deroba, J., Masi, M., **Brodziak, J.**, Aydin, K., Holsman, K., Townsend, H., Tommasi, D., Smith, J., Koenigstein, S., Weijerman, M., Link, J. **2021**. Management strategy evaluation: Allowing the light on the hill to illuminate more than one species. Frontiers in Marine Science, 8:1-21, doi:10.3389/fmars.2021.624355.

Li, B., Shertzer, K., Lynch, K., Ianelli, J., Legault, C., Williams, E., Methot Jr, R., Brooks, E., Deroba, J. Berger, A., Sagarese, S., **Brodziak, J.**, Taylor, I., Karp, M., Wetzel, C., Supernaw, M. **2021**. A comparison of four primary age-structured stock assessment models used in the United States. Fishery Bulletin, 119:149-167, doi:10.7755/FB.119.2-3.5.

Brodziak, J., Sculley, M. **2021**. Observed length composition data for striped marlin, *Kajikia audax*, in the Hawaii longline fishery, 1994-2020. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/21/BILLWG-03/01, 20 p.

Brodziak, J. 2021. Natural Mortality Rates of Pacific Blue Marlin. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/21/BILLWG-01/03, 14 p.

Brodziak, J. 2021. Some Rebuilding Analyses for the Western and Central North Pacific Ocean Striped Marlin Stock. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/21/BILLWG-01/04, 65 p.

Humphreys, R., **Brodziak, J. 2021**. Revised analyses of the reproductive maturity of female striped marlin, Kajikia audax, in the central North Pacific off Hawaii. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/21/BILLWG-02/07, 38 p.

Jardim, E., Azevedo, M., **Brodziak, J.** et al. **2021**. Operationalizing model ensembles for scientific advice to fisheries management. ICES J. Mar Sci. doi:10.1093/icesjms/fsab010

Sculley, M., **Brodziak, J. 2020**. Standardization of Pacific Blue Marlin Catch Per Unit Effort in the Hawaii Longline Fishery from 1995-2019. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/20/BILLWG-03/04, 16 p.

Syslo, J., Brodziak, J., Carvalho, F. **2021**. Stock Assessment Update for the Main Hawaiian Islands Deep 7 Bottomfish Complex in 2021, with Catch Projections through 2025. NOAA Tech. Memo. NMFS-PIFSC -118, 196 p.

Holland, D., Lambert, D., Schnettler, E., Methot, R., Karp, M., Brewster-Geisz, K., **Brodziak, J.**, Crosson, S., Farmer, N., Frens, K., Gasper, J., Hastie, J., Lynch, P., Matson, S., Thunberg, E. **2020**. National Standard 1 Technical Guidance for Designing, Evaluating, and Implementing Carry-over and Phase-in Provisions. NOAA Tech. Memo. NMFS-F/SPO-203, 41 p.

Andrews, A., **Brodziak, J.**, DeMartini, E., Cruz, E. **2020**. A long-lived life history for onaga (*Etelis coruscans*) in the Hawaiian Islands. Mar. Fresh. Res. https://doi.org/10.1071/MF20243

Sculley, M., **Brodziak, J. 2020**. Quantifying the distribution of swordfish (*Xiphias gladius*) density in the Hawaii-based longline fishery. Fisheries Research. https://doi.org/10.1016/j.fishres.2020.105638

Uhrin, A., Walsh, W., **Brodziak, J. 2020**. Relative abundance of derelict fishing gear in the Hawaii-based pelagic longline fishery grounds as estimated from fishery observer data. Scientific Reports.

10:7767. https://doi.org/10.1038/s41598-020-64771-1

Brodziak, J. 2020. On the Probable Distribution of Stock-Recruitment Steepness for Western and Central North Pacific Swordfish. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/20/BILLWG-1/6, 26 p.

Brodziak, J., Sculley, M. **2020**. Which recruitment scenario is most likely for conducting future stock projections of Western and Central North Pacific Ocean striped marlin? International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/20/BILLWG-1/7, 9 p.

Humphreys, R. **Brodziak, J. 2019**. Reproductive maturity of striped marlin, *Kajikia audax*, in the central North Pacific off Hawaii. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/19/BILLWG-2/2, 29 p.

Brooks, E., Thorson, J., Shertzer, K., Nash, R., **Brodziak, J.,** Johnson, K., Klibansky, N., Wells, B., White, J. Paulik revisited: Statistical framework and estimation performance of multistage recruitment functions. **2018**. Fisheries Research. https://doi.org/10.1016/j.fishres.2018.06.018.

Haltuch, M., Brooks, E., **Brodziak, J.**, et al. **2018**. Unraveling the recruitment problem: A review of environmentally-informed forecasting. Fisheries Research. 217:198-216.

Andrews, A., DeMartini, E., **Brodziak, J.**, Nichols, R., Humphreys, R. **2019**. Growth and longevity of Hawaiian grouper (*Hyporthodus quernus*) — input for management and conservation of a large, slow-growing grouper. Canadian Journal of Fisheries and Aquatic Sciences. https://doi.org/10.1139/cjfas-2018-0170.

Lynch, P. D., Methot, R., Link, J, **Brodziak, J**. et al. (eds.). **2018**. Implementing a Next Generation Stock Assessment Enterprise. An Update to the NOAA Fisheries Stock Assessment Improvement Plan. U.S. Dep. Commer., NOAA Tech. Memo. NMFS-F/SPO-183, 127 p. doi: 10.7755/TMSPO.183 Available online at https://spo.nmfs.noaa.gov/content/tech-memo/SAIP2018

Huynh, Q., Gedamke, T., Porch, C., Hoenig, J., Walter, J., Bryan, M., **Brodziak, J. 2017**. Estimating total mortality rates from mean lengths and catch rates in nonequilibrium situations. Transactions of the American Fisheries Society, 146:803-815,DOI: 10.1080/00028487.2017.1308881.

Kapur,M., **Brodziak, J**., Fletcher,E., Yau, A. **2017**. Summary of Life History and Stock Assessment Results for Pacific Blue Marlin, Western and Central North Pacific Striped Marlin, and North Pacific Swordfish. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/17/BILLWG-2/1, 52 p.

Sculley, M., **Brodziak, J.,** Yau, A., Kapur, M. **2017**. An exploratory analysis of trends in swordfish (*Xiphias gladius*) length composition data from the Hawaiian longline fishery. Pacific Islands Fisheries Science Center, PIFSC Working Paper, WP-17-002, 48 p. https://doi.org/10.7289/V5/WP-PIFSC-17-002.

Kai, M., Carvalho, F., Yokoi, H., Kanaiwa, M., Takahashi, N., **Brodziak, J**., Sippel, T., Kohin, S. **2017**. Stock assessment for the north Pacific Blue shark (*Prionace glauca*) using Bayesian State-space Surplus Production Model. ISC/17/SHARKWG-1/4.

Brodziak, J. (presenter), A'mar, T., Supernaw, M. **2017**. Designing a General MSE Framework for a Movement-Based Metapopulation Assessment System. *American Fisheries Society 2017 Annual Meeting Abstracts*, Tampa, Florida.

Haltuch, M., A'mar, T., **Brodziak, J**. (presenter), Brooks, L., Devine, J., Frank, A., Johnson, K.F., Klibansky, N., MacKenzie, B., Nash, R., Payne, M., Shertzer, K., Subbey, S., Wells, B., White, J. **2017**. Unraveling the Recruitment Problem: A Review of Environmentally Informed Forecasting. *American Fisheries Society 2017 Annual Meeting Abstracts*, Tampa, Florida.

Brodziak, J., Supernaw, M. **2017**. Software design document for a metapopulation assessment system. NOAA Technical Report. NMFS Office of Science and Technology, Silver Spring, Maryland.

Carvalho, F., **Brodziak, J. 2016**. Stock assessment and future projections for the North Pacific blue shark (*Prionace glauca*): An alternative Bayesian state-space surplus production model. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/16/SHARKWG-1/17, 26 p.

Walsh, W.A., **Brodziak, J. 2016**. Applications of Hawaii Longline Fishery Observer and Logbook Data for Stock Assessment and Fishery Research. U.S. Dep. Commer., NOAA Tech.

Memo., NOAA-TM-NMFS-PIFSC- 57, 62 p. + Appendices.
Dichmont, C., Deng, R., Punt, A., **Brodziak, J.**, Chang, Y.J., Cope, J., Ianelli, J., Legault, C., Methot, R., Jr., Porch, C., Prager, M., Shertzer, K. **2016**. A review of stock assessment packages in the United States. Fish. Res. 183: 447–460.
Pons, M., Branch, T., Melnychuk, M., Jensen, O., **Brodziak, J.**, Fromentin, J., Harley, S., Haynie, A., Kell, L., Maunder, M., Parma, A., Restrepo, V., Sharma, R., Ahrens, R., Hilborn, R. **2016**. Setting and enforcing catch limits to rebuild depleted tuna and billfish stocks. Fish and Fisheries. http://dx.doi.org/10.1111/faf.12163.
Carvalho, F., **Brodziak, J**. **2016**. Stock assessment and future projections for the North Pacific blue shark (Prionace glauca): An alternative Bayesian State-Space Surplus Production Model. ISC/16/SharkWG-1/17.
Yau, A., Nadon, M., Richards, B., **Brodziak, J.**, Fletcher, E. **2016**. Stock assessment updates of the bottomfish management unit species of American Samoa, the Commonwealth of the Northern Mariana Islands, and Guam in 2015 using data through 2013. U.S. Dept. of Commerce, NOAA Technical Memorandum, NMFS-PIFSC-51, 54 p. doi:10.7289/V5PR7T0G.
Chang, Y.J., Yau, A., **Brodziak, J**. **2016**. Summary of blue marlin (*Makaira nigricans*) catch and size data from the Western and Central Pacific Fisheries Commission and the Inter-American Tropical Tuna Commission. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/16/BILLWG-1/4, 53 p.
Kamikawa, K., Cruz, E., Essington, T., Hospital, J., **Brodziak, J.**, Branch, T. **2015**. Length-weight relationships for 85 fish species from Guam. Journal of Applied Ichthyology, 31:1171-1174.
Brodziak, J. 2015. Changepoint detection for ecosystem processes. *American Fisheries Society 2015 Annual Meeting Abstracts*, Portland, Oregon.
Chang, Y.J., Fletcher, E., Walsh, W., **Brodziak, J**. **2015**. Graphical presentation of the striped marlin (*Kajikia audax*) size composition data to be used in the 2015 stock assessment update. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/15/BILLWG-1/1, 47 p.
Chang, Y.J., Langseth, B., Yau, A., **Brodziak, J**. **2015**. Preliminary stock assessment update for striped marlin (*Kajikia audax*) in the Western and Central North Pacific Ocean through 2013. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/15/BILLWG-1/1, 64 p.
Brodziak, J., Mangel, M., Sun, C. **2015.** Stock-recruitment resilience for North Pacific striped marlin based on reproductive ecology. Fisheries Research, 166:140-150.
Walsh, W., **Brodziak, J**. **2015**. Billfish CPUE standardization in the Hawaii longline fishery: Model selection and multimodel inference. Fisheries Research, 166:151-162.
Chang, Y.J., **J. Brodziak**, J.O'Malley, H.H. Lee, G. DiNardo, C.L. Sun. **2015**. Model selection and multimodel inference for fishery surplus production models: A case study for Pacific blue and striped marlins. Fisheries Research, 166:129-139.
Brodziak, J., Yau, A., O'Malley, J., Andrews, A., Humphreys, R., DeMartini, E., Pan, M., Parke, M., Fletcher, E. **2014**. Stock assessment update for the main Hawaiian Islands Deep7 Bottomfish Complex through 2013 with projected annual catch limits through 2016. U.S. Dept. of Commerce, NOAA Technical Memorandum NOAA-TM-NMFS-PIFSC-42, 61 p.
Belkin, I., G. Hunt Jr., E. Hazen, J. Zamon, R. Schick, R. Prieto, **J. Brodziak**, J. Hare, S. Teo, L. Thorne, H. Bailey, S. Itoh, P. Munk, M. Musyl, J. Willis, W. Zhang. **2014**. Fronts, Fish, and Predators, Deep-Sea Research II, http://dx.doi.org/10.1016/j.dsr2.2014.07.009
Walsh, W., **Brodziak, J**. **2014**. Catch rate standardization for swordfish *Xiphias gladius* in the shallow-set sector of the Hawaii longline fishery, 1995−2012. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/14/BILLWG-1/5, 27 p.
Chang, Y.J., A.J.Y. Yau, **Brodziak, J**. **2014**. Stock assessment of Western and Central North Pacific Ocean swordfish (*Xiphias gladius*) through 2012. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/14/BILLWG-1/2, 47 p.
Yau, A.J.Y., Y.J. Chang, **Brodziak, J**. **2014**. Stock assessment of swordfish (*Xiphias gladius*) in the Eastern Pacific Ocean through 2012. International Scientific Committee for Tuna and Tuna-Like Species in

the North Pacific/Billfish WG, ISC/14/BILLWG-1/1, 54 p.
Brodziak, J., M. Supernaw, and C. Porch. **2014**. Object-Oriented Design of MAS, a Metapopulation Assessment System. A.F.S. *2014 Annual Meeting Abstracts*, Quebec City, Quebec.
Gedamke, T., J. Hoenig, **J. Brodziak**, Q. Huynh, J. Walter, M. Bryan , and C. Porch. **2014**. The future of length-based stock assessments. A.F.S. *2014 Annual Meeting Abstracts*, Quebec City, Quebec.
Brodziak, J. 2014. Abundance Trends and Fisheries for Highly Migratory Species in the Pacific Ocean. A.F.S. *2014 Annual Meeting Abstracts*, Quebec City, Quebec.
W. Walsh and **J. Brodziak**. **2014**. Model Selection and Multimodel Inference for Standardizing Catch Rates of Bycatch Species: A Case Study of Oceanic Whitetip Shark in the Hawaii-Based Longline Fishery. A.F.S. *2014 Annual Meeting Abstracts*, Quebec City, Quebec.
A. Yau, Y. Chang, and **J. Brodziak (presenter). 2014**. Application of Risk Analysis to a Stock Assessment of Swordfish (Xiphias gladius) in the North Pacific Ocean. A.F.S. *2014 Annual Meeting Abstracts*, Quebec City, Quebec.
Brodziak, J., and W. Walsh. **2013.** Model selection and multimodel inference for standardizing catch rates of bycatch species: A case study of oceanic whitetip shark in the Hawaii-based longline fishery. Canadian Journal of Fisheries and Aquatic Sciences 70: 1723–1740 (2013) dx.doi.org/10.1139/cjfas-2013-0111.
Brodziak, J. and E. Fletcher. **2013.** Quarterly summaries of Pacific blue marlin size composition data. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/13/BILLWG-2/1, 278 p.
Brodziak, J. **2013.** A comparison of the consistency of blue marlin (*Makaira nigricans*) length and weight composite data from the Japanese distant water long line fleet. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/13/BILLWG-2/2, 10 p.
Brodziak, J., J. O'Malley, A. Yau, and Y.-J. Chang. **2013.** Application of a Bayesian Production Model to Assess Pacific Blue Marlin (*Makaira nigricans*) in 2013. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/13/BILLWG-2/9, 46 p.
Chang, Y.J., **J. Brodziak**, H.H. Lee, G. Dinardo, and C.L. Sun. **2013**. Model selection uncertainty and multi-model inference in the generalized fishery production modeling: simulation study of the Pacific blue marlin stock. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/13/BILLWG-2/10, 35 p.
Mangel, M., A. MacCall, **J. Brodziak**, E. Dick, R. Forrest, R. Pourzand, and S. Ralston. **2013**. A perspective on steepness, reference points, and stock assessment. Canadian Journal of Fisheries and Aquatic Sciences 70: 1-11. DOI: 10.1139/cjfas-2012-0372.
Chang, S., G. DiNardo, J. Farley, **J. Brodziak**, Z. Yuan. **2013**. Possible stock structure of dolphinfish (*Coryphaena hippurus*) in Taiwan coastal waters and globally based on reviews of growth parameters. Fisheries Research 147: 127-136. DOI: 10.1016/j.fishres.2013.05.003.
Brodziak, J., L. Thomas, and L. Wagatsuma (eds). **2013.** Report of the Billfish Working Group, 16-23 January 2013 NOAA Fisheries Honolulu Service Center, Honolulu, USA, ISC/13/BILLWG-1/REPORT, 46 p.
Brodziak, J. **2013.** Combining information on length-weight relationships for Pacific blue marlin. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/13/BILLWG-1/01, 9 p.
Simon, M., J.-M. Fromentin, S. Bonhommeau, D. Gaertner, **J. Brodziak**, and M.-P. Etienne. **2012**. Effects of stochasticity in early life history on steepness and population growth rate estimates: an illustration on Atlantic bluefin tuna. PLOS ONE 7(10): 1-17.
Brodziak, J., J. O'Malley, B. Richards, and G. DiNardo. **2012**. Stock assessment update of the status of the bottomfish resources of American Samoa, the Commonwealth of the Northern Mariana Islands, and Guam, 2012. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Serv., NOAA, Honolulu, HI 96822-2396. Pacific Islands Fish. Sci. Cent. Admin. Rep. H-12-04, 124 p.
Brodziak, J. **2012**. Fitting Length-Weight Relationships with Linear Regression Using the Log-Transformed Allometric Model with Bias-Correction. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Serv., NOAA, Honolulu, HI 96822-2396. Pacific Islands Fish. Sci. Cent. Admin. Rep. H-12-03, 4 p.
Brodziak, J., X. Dai, and L. Wagatsuma (eds). **2012.** Report of the Billfish Working Group, 2-9 April 2012

Shanghai Ocean University, Shanghai, China, ISC/12/BILLWG-1/REPORT, 68 p.

Brodziak, J., T. Gedamke, C. Porch, J. Walter, D. Courtney, J. O'Malley, and B. Richards. **2012.** A workshop on methods to estimate total and natural mortality rates using mean length observations and life history parameters. U.S. Dep. Commer., NOAA Tech. Memo., NOAA-TM-NMFS-PIFSC-32, 26 p. + Appendix.

Andrews, A. H., R. L. Humphreys, E. E. DeMartini, R. S. Nichols, and **J. Brodziak. 2012.** Comprehensive validation of a long-lived life history for a deep-water snapper (*Pristipomoides filamentosus*) using bomb radiocarbon and lead-radium dating, with daily increment data. Can. J. Fish. Aquat. Sci. 69:1-20. doi:10.1139/f2012-109.

Lee, H.H., K. Piner, R. Humphreys, and **J. Brodziak. 2012.** Stock assessment of striped marlin in the western and central North Pacific Ocean in 2011. ISC/Billfish WG, ISC/SAR/MLS/2012, 119 p. Available at: http://isc.ac.affrc.go.jp/

Link, J., T. Idhe, C. Harvey, S. Gaichas, J. Field, **J. Brodziak**, H. Townsend, and R. Peterman. **2012.** Dealing with uncertainty in ecosystem models: the paradox of use for living marine resource management. Progress in Oceanography, http://dx.doi.org/10.1016/j.pocean.2012.03.008

Brodziak, J., and L. Wagatsuma (eds). **2012.** Report of the Billfish Working Group, 6-16 December 2011 Hawaii Imin International Conference Center, Honolulu, Hawaii, USA, ISC/11/BILLWG-3/REPORT, 53 p.

Brodziak, J., D. Courtney, L. Wagatsuma, J. O'Malley, H.H. Lee, W. Walsh, A. Andrews, R. Humphreys, and G. DiNardo. **2011.** Stock assessment of the main Hawaiian Islands Deep7 bottomfish complex through 2010. U.S. Dep. Commer., NOAA Tech. Memo., NOAA-TM-NMFS-PIFSC-29, 176 p. + Appendix.

Brodziak, J., C.-L. Sun, and L. Wagatsuma (eds). **2011.** Report of the Billfish Working Group, 24 May – 1 June 2011 National Taiwan University, Institute of Oceanography, Chinese Taipei, ROC, 33 p.

Brodziak, J., and M. Mangel. **2011.** Probable values of stock-recruitment steepness for North Pacific striped marlin. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/11/BILLWG-2/11, 13 p.

Brodziak, J., J. Ianelli, K. Lorenzen, and R.D. Methot Jr. (eds). **2011.** Estimating natural mortality in stock assessment applications. U.S. Dep. Commer., NOAA Tech. Memo. NMFS-F/SPO-119, 38 p.

Andrews, A. H., R. L. Humphreys, E. E. DeMartini, R. S. Nichols, and **J. Brodziak**. **2011.** Bomb radiocarbon and lead-radium dating of opakapaka (*Pristipomoides filamentosus*). Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Serv., NOAA, Honolulu, HI 96822-2396. Pacific Islands Fish. Sci. Cent. Admin. Rep. H-11-07, 58 p. + Appendices.

Brodziak, J., and M. Mangel. **2011.** Density-dependent growth and biological reference points for fishery management. A.F.S. *2011 Annual Meeting Abstracts*, Seattle, Washington.

Brodziak, J., H-. H., Lee., and M. Mangel. **2011.** Probable values of stock-recruitment steepness for North Pacific albacore tuna. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Albacore WG, ISC/11/ALBWG-1/11, 10 p.

Ishimura, G., **J. Brodziak**, and M. Kadota. **2011.** Preliminary development of economic capacity and reference points for Western and Central swordfish fisheries (*Xiphias gladius*) in North Pacific. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/11/BILLWG-1/14, 5 p.

Ishimura, G., K. Abe, **J. Brodziak**, and M. Kadota. **2011.** Searching for optimal economic fishing effort for swordfish (*Xiphias gladius*) by Japanese distant water longline fishing vessels. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/11/BILLWG-1/13, 14 p.

Brodziak, J., and G. Ishimura. **2011.** Development of Bayesian surplus production models for assessing the North Pacific swordfish population. Fishery Science 77:22-34.

Brodziak, J., and K. Piner. **2010.** Model averaging and probable status of North Pacific striped marlin, *Tetrapturus audax*. Can. J. Fish. Aquat. Sci. 67: 793-805.

Ichinokawa, M. and **J. Brodziak. 2010.** Using adaptive area stratification to standardize catch rates with application to North Pacific swordfish (*Xiphias gladius*). Fish. Res. 106:249:260.

Mangel, M., **J. Brodziak**, and G. DiNardo. **2010.** Accounting for variation in mortality and Allee effects when

computing steepness for strategic fisheries management. University of California, Santa Cruz, Jack Baskin School of Engineering Technical Report, UCSC-SOE-10-33, 41 p.

Brodziak, J., M. Mangel, and G. DiNardo. **2010.** Estimating stock-recruitment steepness from life history information: a case study of North Pacific bluefin tuna, *Thunnus orientalis*. A.F.S. *2010 Annual Meeting Abstracts*, Pittsburgh, Pennsylvania.

Brodziak, J., and D. Courtney. **2010.** Age-based analyses of potential biological reference points for the Western and Central North Pacific swordfish (*Xiphias gladius*) stock. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/10/BILLWG-2/02, 18 p.

Brodziak, J., and G. Ishimura. **2010.** Production model analyses of maximum sustainable yield-based reference points for the North Pacific swordfish stocks. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/10/BILLWG-2/03, 4 p.

Brodziak, J., and G. Ishimura. **2010.** Stock assessment of North Pacific swordfish (*Xiphius gladius*) in 2009. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Ser., NOAA, Honolulu, HI 96822-2326. Pacific Islands Fish. Sci. Cent. Admin. Rep. H-10-01.

Brodziak, J., K. Piner, and D. Courtney. **2010.** Modeling recruitment responses of striped marlin (*Tetrapturus audax*) and swordfish (*Xiphias gladius*) to environmental variability in the North Pacific. *Climate Change Effects on Fish and Fisheries. PICES International Symposium Program and Abstracts*, Sendai, Japan.

Brodziak, J. 2010. Update of the production model assessment of the eastern Pacific swordfish stock (*Xiphias gladius*) in 2010. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/10/BILLWG-1/02, 31 p.

Brodziak, J. 2010. An investigation of patterns in Japanese longline CPUE of striped marlin (*Tetrapturus audax*) in relation to a stock boundary of swordfish (*Xiphias gladius*) in the North Pacific. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/10/BILLWG-1/06, 19 p.

Courtney, D., and **J. Brodziak. 2010.** Oceanographic features in the vicinity of a North Pacific swordfish stock International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/09/BILLWG-2/04, 16 p.

Mangel, M., **J. Brodziak**, and G. DiNardo. **2010.** Reproductive ecology and scientific inference of steepness. Fish and Fisheries. 11:89-104.

Brodziak, J., R. Moffitt, and G. DiNardo. **2009.** Hawaiian bottomfish assessment update for 2008. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Ser., NOAA, Honolulu, HI 96822-2326. Pacific Islands Fish. Sci. Cent. Admin Rep. H-09-02, 93 p.

Courtney, D., W. Walsh, and **J. Brodziak. 2009.** Generalized additive model analyses to standardize swordfish (*Xiphias gladius*) catch rates in the Hawaii-based pelagic longline fishery, 1995-2007, for use in stock assessment. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/09/BILLWG-1/04, 33 p.

Brodziak, J. 2009. Fish stocks/Overfishing. In R. Gillespie and D. Clague (eds.), *Encyclopedia of Islands*, pp. 310-311, University of California Press, Berkeley, CA.

Brodziak, J., and G. Ishimura. **2009.** Development of Bayesian surplus production models for assessing the North Pacific swordfish population. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/09/BILLWG-2/02, 29 p.

Ishimura,G., and J. Brodziak. **2009.** Model-averaging to account for prior uncertainty in swordfish intrinsic growth rate and carrying capacity. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/09/BILLWG-2/03, 14 p.

Brodziak, J., and K. Piner. **2009.** Model Uncertainty and biological reference points for North Pacific striped marlin. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/09/BILLWG-1/02, 32 p.

Friedland, K., J. Hare, G. Wood, L. Col, L. Buckley, D. Mountain, J. Kane, **J. Brodziak**, G. Lough, and C. Pilskaln. **2009.** Reply to the comment by Payne et al. on “Does the fall phytoplankton bloom control recruitment of Georges Bank haddock, *Melanogrammus aeglefinus*, through parental condition?” Can. J. Fish. Aquat. Sci. 65:1076-1086.

Brodziak, J., and D. Courtney. **2009.** Length distributions of female and male swordfish, *Xiphias gladius*, captured in the directed Hawaii pelagic longline fishery during 1994-2008. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/09/BILLWG-1/07, 9 p.

Brodziak, J. 2009. Potential natural mortality rates of North Pacific swordfish. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/09/BILLWG-1/13, 20 p.

Brodziak, J., S.X. Cadrin, C.M Legault, and S.A. Murawski. **2008.** Goals and strategies for rebuilding New England groundfish stocks. Fish. Res. 94:355-366.

Brodziak, J., M. Traver, and L. Col. **2008**. The nascent recovery of the Georges Bank haddock stock. Fish. Res. 94:123-132.

Ichinokawa, M., and **J. Brodziak. 2008.** Possible substock boundaries of swordfish, *Xiphias gladius*, population in the Pacific Ocean judged from annual CPUE trends by Japanese longliners. Proceedings of the Fifth World Fisheries Congress, Yokohama, Japan, 2 p.

Ichinokawa, M., and **J. Brodziak. 2008.** Stock boundary between possible swordfish stocks in the northwest and southeast Pacific judged from fisheries data of Japanese longliners. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/08/Special Session on Billfish Stock Structure/06, 14 p.

Grober-Dunsmore, R., L. Wooninck, J. Field, C. Ainsworth, J. Beets, S. Berkeley, J. Bohnsack, R. Boulon, R. Brodeur, **J. Brodziak**, L. Crowder, D. Gleason, M. Hixon, L. Kaufman, B. Lindberg, M. Miller, L. Morgan, and C. Wahle. **2008.** Vertical zoning in marine protected areas: Ecological considerations for balancing pelagic fishing with conservation of benthic communities. Fisheries. 33: 598-610.

Brodziak, J., and K. Piner. **2008.** Maximum sustainable yield-based reference points for North Pacific striped marlin. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/08/BILLWG-2/09, 13 p.

Brodziak, J. 2008. An Investigation of length-based estimates of total mortality for North Pacific striped marlin, *Tetrapturus audax*. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/08/BILLWG-2/10, 10 p.

Courtney, D., W. Walsh, and **J. Brodziak. 2008.** Generalized additive model analyses to standardize swordfish (*Xiphias gladius*) catch rates in the Hawaii-based longline fishery, 1995 – 2006. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/08/BILLWG-2/08, 10 p.

Brodziak, J. 2008. An assessment of the risk of archipelagic overfishing for alternative total allowable catches of deep-7 bottomfish in the main Hawaiian Islands. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Serv., NOAA, Honolulu, HI 96822-2396. Pacific Islands Fish. Sci. Cent. Admin. Rep. H-08-03, 31 p.

McCarthy, J., K. Friedland, and **J. Brodziak. 2008.** Enhancement of image-based fecundity methods: Gravimetric sampling at sea and safer sample preservation. *Fish. Res.* 93:47-53.

Brodziak, J., and K. Piner. **2008.** Biological reference points and status of North Pacific striped marlin, *Tetrapturus audax*. A.F.S. *2008 Annual Meeting Abstracts*, Ottawa, Canada.

Brodziak, J., and W. Walsh. **2008.** Model selection for standardizing striped marlin catch-per-unit effort in the Hawaii-based longline fishery. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/Billfish WG, ISC/08/BILLWG-1/01, 18 p.

Friedland, K., J. Hare, G. Wood, L. Col, L. Buckley, D. Mountain, J. Kane, **J. Brodziak**, G. Lough, and C. Pilskaln. **2008.** Does the fall phytoplankton bloom control recruitment of Georges Bank haddock, *Melanogrammus aeglefinus*, through parental condition? *Can. J. Fish. Aquat. Sci*. 65:1076-1086.

Link, J., W. Overholtz, J. O'Reilly, J. Green, D. Dow, D. Palka, C. Legault, J. Vitaliano, V. Guida, M. Fogarty, **J. Brodziak**, L. Methratta, W. Stockhausen, L. Col and C. Griswold. **2008.** The Northeast U.S. continental shelf energy modeling and analysis exercise (EMAX): Ecological network model development and basic ecosystem metrics. J. Mar. Sys. 74:453474.

Brodziak, J., and J. Link. **2008**. The incredible shrinking Georges Bank haddock (*Melanogrammus aeglefinus*). In G.Kruse, K. Drinkwater, J. Ianelli, J. Link, D. Stram, V. Wespestad, and D. Woodby (eds.), *Resiliency of gadid stocks to fishing and climate change*, pp. 141-160, Alaska Sea Grant

College Program, AK-SG-08-01, Univ. of Alaska, Fairbanks.
Link, J.S., O'Reilly, J., Dow, D., Fogarty, M., Vitaliano, J., Legault, C., Overholtz, W., Green, J., Palka, D., Guida, V., and **Brodziak, J. 2007.** Comparisons of the Georges Bank Ecological Network: EMAX in Historical Context. J. Northwest Atl. Fish. Sci. 39:83-101.
Moffitt, R., **J. Brodziak**, and T. Flores. **2007.** Status of the Bottomfish Resources of American Samoa, Guam, and Commonwealth of the Northern Mariana Islands, 2005. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Ser., NOAA, Honolulu, HI 96822-2326. Pacific Islands Fish. Sci. Cent. Admin Rep. H-07-04, 52 p.
Brodziak, J.K.T., S.X. Cadrin, C.M Legault, and S.A. Murawski. **2007.** Goals and strategies for rebuilding New England groundfish stocks. A.F.S. *2007 Annual Meeting Abstracts*, San Francisco, CA.
Brodziak, J., and G. DiNardo. **2007.** Using probability logic to guide harvest policy development and implementation. A.F.S. *2007 Annual Meeting Abstracts*, San Francisco, CA.
Brodziak, J. 2007. Preliminary calculations of yield and spawning biomass per recruit biological reference points for striped marlin. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/MARWG-SWORWG-2/Working Paper 3.
Ichinokawa, M. and **J. Brodziak. 2007.** Area stratification using the tree-glm algorithm: A new method for standardizing CPUE of swordfish in the North Pacific. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/MARWG-SWORWG-2/Working Paper 2.
Piner, K., R. Conser, G. Dinardo, and **J. Brodziak. 2007.** Evaluation of model performance from the 2007 ISC striped marlin stock assessment. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/MARWG-SWORWG-1/Working Paper 4.
Brodziak, J., R. Moffitt, and G. Dinardo. **2007.** Main Hawaiian Islands Total Allowable Catch Alternatives. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Ser., NOAA, Honolulu, HI 96822-2326. Pacific Islands Fish. Sci. Cent. Internal Rep. H-07-010, 6 p.
Brodziak, J. 2007. An investigation of alternative production models to assess the Hawaiian bottomfish complex. Pacific Islands Fish. Sci. Cent., Natl. Mar. Fish. Ser., NOAA, Honolulu, HI 96822-2326. Pacific Islands Fish. Sci. Cent. Admin. Rep. H-07-01, 71 p.
Mayo, R.M., **J. Brodziak**, J. Burnett, M. Traver, and L. Col. **2007.** The 2005 assessment of Acadian redfish, *Sebastes faciatus*, in the Gulf of Maine/Georges Bank region. *NEFSC Ref. Doc.* 07-06. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0706/
Brodziak, J. 2007. Using the EM algorithm to predict catch-per-unit effort of striped marlin by the Japanese distant water longline fleet in Region 5. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/MARWG-SWORWG-1/Working Paper 1, 6 p.
Piner, K., R. Conser, G. Dinardo, and **J. Brodziak. 2007.** Stock synthesis 2 sensitivity runs for striped marlin assessment WG 2007. International Scientific Committee for Tuna and Tuna-Like Species in the North Pacific/MARWG-SWORWG-1/Working Paper 2.
Brodziak, J., and W. Walsh. **2007.** Aspects of model selection for generalized linear models applied to striped marlin in the Hawaii-based longline fishery. Pelagic Longline Catch Rate Standardizations Workshop, Honolulu, HI. Working Paper 1, 68 p.
Gavaris, S., R. Mayo, **J. Brodziak**, J. Hunt, C. Legault, L. O'Brien, H. Stone, and L. Van Eeckhaute. **2006.** Length Composition Comparisons of USA and Canadian Fishery Catches for Atlantic Cod, Haddock and Yellowtail Flounder. TRAC Reference Document 2006/02. Available at: http://www.mar.dfo-mpo.gc.ca/science/TRAC/rd.html
Brodziak, J., and J. Link. **2006.** The incredible shrinking Georges Bank haddock (*Melanogrammus aeglefinus*) [abstract]. Presented at: 24th Wakefield Symposium. Resiliency of Gadid Stocks to Fishing and Climate Change: 2006 Oct 31 - Nov 3; Anchorage, AK.
Brodziak, J., M. Traver, L. Col, and S. Sutherland. **2006.** Stock assessment of Georges Bank haddock, 1931-2004. *NEFSC Ref. Doc.* 06-11. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0611/
Thunberg, E., C. Fulcher, and **J. Brodziak. 2006.** Harvest strategies for a transboundary resource: Georges Bank haddock. Available from: http://oregonstate.edu/dept/iifet/. In: Shriver AL, compiler. Proceedings of the Thirteenth Biennial Conference of the International Institute of Fisheries Economics & Trade, July 11-14, 2006, Portsmouth, UK: Rebuilding Fisheries in an Uncertain

Environment [CD-ROM]. Corvallis (OR): IIFET.
Link, J., J. O'Reilly, D. Dow, M. Fogarty, J. Vitaliano, C. Legault, W. Overholtz, J. Green, D. Palka, V. Guida, and **J. Brodziak. 2006.** Comparisons of the Georges Bank ecological network: EMAX in historical context [abstract]. Presented at: Northwest Atlantic Fisheries Organization [NAFO] Science Symposium: Environmental and Ecosystem Histories in the Northwest Atlantic - What Influences Living Marine Resources?; 2006 Sep 13-15; Dartmouth, NS
Brodziak, J., and S. Correia. **2006.** Classic examples of recovering stocks: Achieving success by learning from the past in New England fisheries. A.F.S. *2006 Annual Meeting Abstracts*, Lake Placid, New York.
Col, L., and **J. Brodziak. 2006.** Using a traffic light approach to determine the stock status of Atlantic halibut and Atlantic wolffish. A.F.S. *2006 Annual Meeting Abstracts*, Lake Placid, New York.
Rago, P., **J. Brodziak**, and C. Legault. **2006.** Changes in data and dynamics during rebuilding: the assessment challenges of rapidly recovering stocks. A.F.S. *2006 Annual Meeting Abstracts*, Lake Placid, New York.
Sigourney, D., M. Ross, **J. Brodziak**, and J. Burnett. **2006.** Growth, sexual maturity and distribution of the Atlantic halibut in the Georges Bank-Gulf of Maine region. *Journal of Northwest Atlantic Fisheries Science*. V36, art. 7. Available at: http://journal.nafo.int/36/sigourney/7-sigourney.html
Thunberg, E., D. Liew, C. Fulcher, and **J. Brodziak. 2005.** Economic analysis of alternative harvest strategies for eastern Georges Bank haddock. *NEFSC Ref. Doc.* 05-15. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0515/
Brodziak, J., and L. O'Brien. **2005.** Do environmental factors affect recruits per spawner anomalies of New England groundfish? *ICES J. Mar. Sci.* 62:1394-1407.
Pershing, A., C. Greene, J. Jossi, L. O'Brien, **J. Brodziak**, and B. Bailey. **2005.** Interdecadal variability in the Gulf of Maine zooplankton community with potential impacts on fish recruitment. *ICES J. Mar. Sci.* 62:1511-1523.
Van Eeckhaute, L. and **J. Brodziak. 2005**. Assessment of Eastern Georges Bank Haddock. Transboundary Resource Assessment Committee Ref. Doc. 2005/03. 73 p. Available at: http://www.mar.dfo-mpo.gc.ca/science/TRAC/rd.html
Brodziak, J., M. Traver, and L. Col. 2005. Georges Bank haddock. In NEFSC, *Assessment of 19 northeast groundfish stocks through 2004*, pp. 2:30-80. *NEFSC Ref. Doc.* 05-13, 499 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0513/
Brodziak, J., and M. Traver. **2005.** Gulf of Maine haddock. In NEFSC, *Assessment of 19 northeast groundfish stocks through 2004*, pp. 2:415-423. *NEFSC Ref. Doc.* 05-13, 499 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0513/
Brodziak, J., and L. Col. **2005.** Atlantic halibut. In NEFSC, *Assessment of 19 northeast groundfish stocks through 2004*, pp. 2:424-431. *NEFSC Ref.Doc.*05-13, 499 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0513/
Brodziak, J. 2005. Maximum likelihood estimators of stock composition. In S. Cadrin, K. Friedland, and J. Waldman (Eds.), *Stock Identification Methods,* Elsevier, New York, pp. 571-589.
Brodziak, J. K. T., and C. Legault. **2005.** Model averaging to estimate rebuilding targets for overfished stocks. *Can. J. Fish. Aquat. Sci.* 62:544-562.
Brodziak, J. 2005. Essential fish habitat source document: Haddock, *Melanogrammus aeglefinus*, life history and habitat characteristics, 2nd edition. NOAA Tech Mem. NMFS-NE-196, 74 pp. Available at: http://www.nefsc.noaa.gov/nefsc/publications/tm/tm196/
Link, J., W. Overholtz, J. O'Reilly, J. Green, E. Methratta, D. Dow, D. Palka, C. Legault, S. Edwards, G. Waring, W. Stockhausen, D. Mountain, J. Vitaliano, V. Guida, J. Kane, J. Jossi, M. Fogarty, **J. Brodziak**, C. Griswold, C. McCandless, N. Kohler, S. Fromm, L. Col, T. Smith, C. MacKenzie, and R. Goldberg. **2005.** An Overview of EMAX: The Northeast U.S. Continental Shelf Ecological Network. *ICES [International Council for the Exploration of the Sea] C.M. 2005. L:02, 40 pp.*
Link, J., Almeida, F., Reid, B., Packer, D., Vitaliano, J., Noji, T., Murawski, S., McMillan , S., Brown, R., Rago, P., Guida, V., Collie, J., Valentine, P., Auster, P., Lindholm, J., Methratta, L., **Brodziak, J.**, Arlen, L., Fogarty, M. **2005**. What We Have Learned from Large Scale Closed Areas: the

Northeast U.S. Example. The First International Marine Protected Areas Congress (IMPAC1) Conference Proceedings.
Link, J., M. Fogarty, **J. Brodziak**, W. Overholtz, T. Noji, S. Murawski, and F. Serchuk. **2005.** A case history of ecosystem considerations for science supporting fishery management in the Northeast US. *ICES C.M. 2005/BB:02.*
Brodziak, J., C. Legault, L. Col, and W. Overholtz. **2005.** Trends in demersal and pelagic species biomasses in the northeast USA continental shelf ecosystem. A.F.S. *2005 Annual Meeting Abstracts*, Anchorage, Alaska.
Traver, M., L. Col, and **J. Brodziak. 2005.** A preliminary look at the Georges Bank haddock 2003 baby boom. A.F.S. *2005 Annual Meeting Abstracts*, Anchorage, Alaska.
Brodziak, J.K.T., S.X. Cadrin, C.M Legault, and S.A. Murawski. **2004.** Goals and strategies for rebuilding New England groundfish stocks. *ICES C.M. 2004. K:42, 32 p.*
Brodziak, J.K.T., C.M. Legault, L.A. Col, and W.J. Overholtz. **2004.** Estimation of demersal and pelagic species biomasses in the northeast USA continental shelf ecosystem. *ICES C.M. 2004. FF:07, 31p.*
Brodziak, J.K.T., P. M. Mace, W. J. Overholtz, and P. J. Rago. **2004.** Ecosystem trade-offs in managing New England fisheries. *Bull. Mar. Sci.* 74(3):529-548.
Van Eeckhaute, L., and **J. Brodziak. 2004.** Assessment of eastern Georges Bank haddock. Canadian Science Advisory Secretariat Research Document 2004/076. Available at: http://www.mar.dfo-mpo.gc.ca/science/TRAC/rd.html
Brodziak, J.K.T., and Link, J.S. **2003.** An overview of metrics reflecting ecosystem status. In: Proceedings of the Seventh NMFS National Stock Assessment Workshop: (Re)building Sustainable Fisheries and Marine Ecosystems. NOAA Tech. Mem. NMFS.
Link, J.S. and **Brodziak, J.K.T.** [Rapporteurs]. **2003.** Topic 1: Defining "ecosystem overfishing." [Discussion group report]. Proceedings of the Seventh NMFS National Stock Assessment Workshop. NOAA Tech. Mem. NMFS-F/SPO.
Wigley, S., **J. K. T. Brodziak**, and L. Col. **2003.** Assessment of the Gulf of Maine and Georges Bank Witch Flounder Stock for 2003. *NEFSC Ref. Doc.* 03-14, 194 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0314/
Brodziak, J. K. T. 2003. Three approaches for addressing uncertainty in stock composition analysis. *ICES C.M. 2003. V20. 16 p.*
Van Eeckhaute, L., S. Gavaris, and **J. Brodziak. 2003.** Assessment of haddock on eastern Georges Bank. Canadian Science Advisory Secretariat Research Document 2003/076. Available at: http://www.dfo-mpo.gc.ca/csas/
Brodziak, J., and C. Legault. **2003.** What are appropriate rebuilding targets for overfished stocks? A.F.S. *2003 Annual Meeting Abstracts*, Quebec City, Quebec.
Brodziak, J.K.T., and D. G. Mountain. **2002.** Some biological responses of fish stocks to increased shelf water volume in the Mid-Atlantic Bight of the northeast USA continental shelf ecosystem. *ICES C.M. 2002. Q4. 31 pp.*
Brodziak, J., M. Thompson, and R. Brown. **2002.** Georges Bank haddock. In NEFSC, *Assessment of 20 northeast groundfish stocks through 2001*, pp. 36-59. *NEFSC Ref. Doc.* 02-16, 509 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0216/
Brodziak, J., and M. Thompson. **2002.** Gulf of Maine haddock. In NEFSC, *Assessment of 20 northeast groundfish stocks through 2001*, pp. 298-305. *NEFSC Ref. Doc.* 02-16, 509 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0216/
Brodziak, J. 2002. Atlantic halibut. In NEFSC, *Assessment of 20 northeast groundfish stocks through 2001*, pp.306-313. *NEFSC Ref. Doc.* 02-16, 509 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0216/
Link, J. S., and **J.K.T. Brodziak** (eds). **2002.** Status of the Northeast U.S. Continental Shelf Ecosystem. *NEFSC Ref. Doc.* 02-11, 254 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0211/
Link, J., **J. Brodziak**, S. Edwards, W. Overholtz, D. Mountain, J. Jossi, T. Smith, and M. Fogarty. **2002.**

Marine ecosystem assessment in a fisheries management context. *Can. J. Fish. Aquat. Sci.* 59:1429-1440.

Brodziak, J., and J. Link. **2002.** Ecosystem-based fishery management: What is it and how can we do it? *Bull. Mar. Sci.* 70(2):589-611.

Brodziak, J.K.T., W. J. Overholtz, and P. J. Rago. **2002.** Rejoinder: Interpreting spawning stock and recruitment data in New England groundfish. *Can. J. Fish. Aquat. Sci.* 59:193-195.

Brodziak, J. 2002. In search of optimal harvest rates for west coast groundfish. *N. Amer. J. Fish. Mgt.* 22:258-271.

Murawski, S., M. Fogarty, P. Rago, and **J. Brodziak. 2002.** Quantitative methods for MPA design, with application to the NE USA. In L. Madin (Ed.), *Marine protected areas: Design and implementation for conservation and fisheries restoration*, pp. 41-44. Woods Hole Oceanographic Institution, Woods Hole, MA.

Brodziak, J. 2002. An age-structured assessment model for Georges Bank winter flounder. NEFSC Ref. Doc. 02-03, 54 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0203/

Mayo, R. K., **J.K.T. Brodziak**, M. Thompson, J. Burnett, and S.X. Cadrin. **2002.** Biological Characteristics, Population Dynamics, and Current Status of Redfish, *Sebastes fasciatus Storer*, in the Gulf of Maine - Georges Bank Region. *NEFSC Ref. Doc.* 02-05, 138 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0205/

Link, J., **J. Brodziak**, S. Edwards, W. Overholtz, D. Mountain, J. Jossi, T. Smith, and M. Fogarty. **2001.** Ecosystem status in the northeast United States continental shelf ecosystem: Integration, synthesis, trends, and the meaning of ecosystem metrics. *ICES C.M. 2001. T10. 41 p.*

Macy, W.K., III., and **J. Brodziak. 2001.** Seasonal maturity and size at age of *Loligo pealeii* in waters of southern New England. *ICES J. Mar. Sci.* 58: 852-864.

Jacobson, L.D., **J. Brodziak**, and J. Rogers. **2001.** Depth distributions and time varying bottom trawl selectivities for Dover sole (*Microstomus pacificus*), sablefish (*Anoplopoma fimbia*) and thornyheads (*Sebastolobus alascanus* and *S. altivelis*) in a commercial fishery. *Fish. Bull. U.S.* 99:309-327.

Brodziak, J.K.T., W. J. Overholtz, and P. J. Rago. **2001.** Does spawning stock affect recruitment of New England groundfish? *Can. J. Fish. Aquat. Sci.* 58:306-318.

Brodziak, J., E. Holmes, K. Sosebee, and R. Mayo. **2001.** Assessment of the silver hake resource in the northwest Atlantic in 2000. *NEFSC Ref. Doc.* 01-03, 134 p. Available at: http://www.nefsc.noaa.gov/nefsc/publications/crd/crd0103/

Brodziak, J., and R. Mikus. **2000.** Variation in the life history parameters of Dover sole, *Microstomus pacificus*, off the coasts of Washington, Oregon, and California. *Fish. Bull., U.S.* 98:661-673.

Brodziak, J., and J. Link. **2000.** Ecosystem management: How can we do it? In P. Mace. (Ed.), *Proceedings of the Sixth NMFS National Stock Assessment Workshop*. NOAA Tech. Mem. NMFS-F/SPO-46: 16.

Wigley, S., J. Brodziak, and S. Cadrin. **1999.** Assessment of the witch flounder stock in subareas 5 and 6 for 1999. *NEFSC Ref. Doc.* 99-16, 153 p.

Brodziak, J., and L. Hendrickson. **1999.** An analysis of environmental effects on survey catches of squids, *Loligo pealei* and *Illex illecebrosus*, in the northwest Atlantic. *Fish. Bull., U.S.* 97:9-24.

Overholtz, W., S.A. Murawski, P.J. Rago, W.L. Gabriel, M. Terceiro, and **J.K.T. Brodziak. 1999.** Ten-year projections of landings, spawning stock biomass, and recruitment for five New England groundfish stocks. *NEFSC Ref. Doc.* 99-05. Woods Hole, MA, 74 pp.

Brodziak, J., P. Rago, and R. Conser. **1998.** A general approach for making short-term stochastic projections from an age-structured fisheries assessment model. In F. Funk, T. Quinn II, J. Heifetz, J. Ianelli, J. Powers, J. Schweigert, P. Sullivan, and C.-I. Zhang (eds), *Proceedings of the International Symposium on Fishery Stock Assessment Models for the 21st Century*. Alaska Sea Grant College Program, Univ. of Alaska, Fairbanks.

Brodziak, J. 1998. Revised biology and management of long-finned squid, *Loligo pealei*, in the northwest Atlantic. *CalCOFI Rep.* 39:61-70.

Rago, P.J., K.A. Sosebee, **J.K.T. Brodziak**, S.A. Murawski, and E.D. Anderson. **1998.** Implications of recent increases in catchs on the dynamics of Northwest Atlantic spiny dogfish (*Squalus*

acanthias). Fish. Res. 39:165-181.
Lipinski, M. R., C. J. Augustyn, D. S. Butterworth, **J.K.T. Brodziak**, G. Christy, S. des Clers, G.D. Jackson, R. K. O'Dor, D. Pauly, L.V. Purchase, M. J. Roberts, B. A. Roel, Y. Sakurai, and W. H. H Sauer. **1998.** Cephalopod fisheries: a future global upside to past overexploitation of living marine resources? *In:* A. Payne, M. Lipinski, M. Clarke, and M. Roeleveld (eds). S. Afr. J. Mar. Sci. 20:463-469.
Dawe, E. G. and **J.K.T. Brodziak. 1998.** Trophic relationships, ecosytem variability, and recruitment. *In*: Rodhouse, P.G.; Dawe, E.G.; O'Dor, R.K. (eds.) Squid recruitment dynamics: The genus Illex as a model, the commercial Illex species and influences on variability. FAO Fisheries Technical Paper. No. 376. Rome, FAO. 1998. 273p.
Helser, T.E. and **J.K.T. Brodziak. 1998.** Impact of density-dependent growth and maturation on assessment advice to rebuild depleted U.S. silver hake (*Merluccius bilinearis*) stocks. *Can. J. Fish. Aquat. Sci*. 55:882-892.
Rogers, J., T. Builder, P. Crone, **J. Brodziak**, R. Methot, R. Conser, and R. Lauth. **1998.** Status of the shortspine thornyhead (*Sebastolobus alascanus*) resource in 1998. *In*: Pacific Fishery Management Council. *Appendix: Status of the Pacific coast groundfish fishery through 1998 and recommended biological catches for 1999: Stock assessment and fishery evaluation*. Pacific Fishery Management Council, 2130 SW Fifth Avenue, Suite 224, Portland, Oregon 97201.
Methot, R., P. Crone, R. Conser, **J. Brodziak**, T. Builder, and D. Kamikawa. **1998.** Status of the sablefish resource off the U.S. Pacific coast in 1998. *In*: Pacific Fishery Management Council. *Appendix: Status of the Pacific coast groundfish fishery through 1998 and recommended biological catches for 1999: Stock assessment and fishery evaluation*. Pacific Fishery Management Council, 2130 SW Fifth Avenue, Suite 224, Portland, Oregon 97201.
Brodziak, J. 1997. Standardized catch rates for the deep-water complex. Division Research Report. NMFS, Northwest Fisheries Science Center, Fishery Resource Analysis and Monitoring Division, Hatfield Marine Science Center, Newport, Oregon. 49 p.
Brodziak, J., L. Jacobson, R. Lauth, and M. Wilkins. **1997.** Assessment of the Dover sole stock for 1997. *In*: Pacific Fishery Management Council. *Appendix: Status of the Pacific coast groundfish fishery through 1997 and recommended biological catches for 1998: Stock assessment and fishery evaluation*. Pacific Fishery Management Council, 2130 SW Fifth Avenue, Suite 224, Portland, Oregon 97201.
Hayes, D.B., and **J.K.T. Brodziak. 1997.** Reply: Efficiency and bias of estimators and sampling designs for determining length-weight relationships of fish. *Can. J. Fish. Aquat. Sci.* 54: 744-745.
Brodziak, J.K.T., and L.C. Hendrickson. **1997.** An analysis of some factors affecting survey catches of squids, *Loligo pealeii* and *Illex illecebrosus*, in the northwest Atlantic. *NEFSC Ref. Doc.* 97-03. Woods Hole, MA.
Erickson, D., E. Pikitch, R. Methot, E. Babcock, T. Builder, J. Wallace, P. Crone, **J. Brodziak**, and R. Conser. **1997.** Survival of trawl caught and discarded sablefish (*Anoplopoma fimbria*) off the U.S. west coast. *ICES C.M. 1997*/FF:06.
Brodziak, J.K.T., and W. K. Macy, III. **1996.** Growth of long-finned squid, *Loligo pealei*, in the northwest Atlantic. *Fish. Bull., U.S.* 94:212-236.
Brodziak, J., R. Conser, L. Jacobson, T. Jaglielo, and G. Sylvia. **1996.** Groundfish stock assessment review meeting - June 3-7, 1996 in Newport, Oregon. *In:* Status of the Pacific coast groundfish fishery through 1996 and recommended acceptable biological catches for 1997. Pacific Fishery Management Council, 2130 SW Fifth Ave, Suite 224, Portland, Oregon, USA.
Helser, T. E. and **J.K.T. Brodziak. 1996.** Influence of temperature and depth on distribution and catches of yellowtail flounder, Atlantic cod, and haddock in NEFSC bottom trawl surveys. *NEFSC Ref. Doc.* 96-05e. Woods Hole, MA.
Hendrickson, L.C., **J. Brodziak**, M. Basson, and P. Rago. **1996.** Stock assessment of the northern shortfin squid in the northwest Atlantic during 1993. *NEFSC Ref. Doc.* 96-05g. Woods Hole, MA.
Overholtz, W. J., S. F. Edwards, and **J.K.T. Brodziak. 1995.** Effort control in the New England groundfish fishery: A bioeconomic perspective. *Can. J. Fish. Aquat. Sci.* 52:1944-1957.
Hayes, D. B., **J.K.T. Brodziak**, and J. B. O'Gorman. **1995.** Efficiency of estimators and sampling designs

for determining length-weight relationships. *Can J. Fish. Aquat. Sci.* 52:84-92
Brodziak, J., and W. Ling. **1995.** An examination of the influence of environmental conditions on spring survey catches of Atlantic mackerel. *NEFSC Ref. Doc.* 95-14.Woods Hole, MA
Weinberg, R. J., S. A. Murawski, R. Conser, **J. Brodziak**, L. Hendrickson, H. Lai, and P. Rago. **1995.** Analytical assessment of surfclam populations in the Middle Atlantic region of the United States in 1994. *NEFSC Ref. Doc.* 95-05. Woods Hole, MA
Murawski, S., J. Weinberg, P. Rago, **J. Brodziak**, L. Hendrickson, R. Conser, and H. Lai. **1995.** Ocean quahog populations from the Middle Atlantic to the Gulf of Maine in 1994. *NEFSC Ref. Doc.* 95-07. Woods Hole, MA.
Helser, T.E., and **J.K.T. Brodziak. 1995.** Impact of compensatory responses on assessment advice concerning overfishing targets and rebuilding of a depleted fish stock. *A.F.S. 1995 Annual Meeting Abstracts,* Tampa, Florida, p. 23.
Rago, P. J., K. Sosebee, **J.K.T. Brodziak**, and E. Anderson. **1995.** Distribution and dynamics of northwest Atlantic spiny dogfish *Squalus acanthias*. *A.F.S. 1995 Annual Meeting Abstracts*, Tampa, Florida, p. 132.
Brodziak, J.K.T., and W. K. Macy, III. **1994.** Revised estimates of growth of long-finned squid, *Loligo pealei*, in the Northwest Atlantic based on statolith ageing: Implications for stocks assessment and fishery management. *ICES C.M. 1994*/K:13.
Fogarty, M.J., and **J.K.T. Brodziak. 1994.** Multivariate time series analysis of fish community dynamics on Georges Bank. *ICES C.M. 1994*/P:3.
Brodziak, J., and P. J. Rago. **1994.** A general approach for short-term stochastic projections in age-structured fisheries assessment models. *Working Paper #4. SARC Assessment Subcommittee Meeting*. May 4-6, 1994. NEFSC, Woods Hole, MA.
Brodziak, J. 1994. Stock assessment for Atlantic butterfish, *Peprilus triacanthus*, in the Northwest Atlantic during 1992. *NEFSC Ref. Doc.* 94-03. Woods Hole, MA.
Brodziak, J. 1994. Stock assessment for long-finned squid, *Loligo pealei*, in the Northwest Atlantic during 1992. *NEFSC Ref. Doc*. 94-04. Woods Hole, MA.
Brodziak, J., and L. Hendrickson. **1994.** Stock assessment for short-finned squid, *Illex illecebrosus*, in the Northwest Atlantic during 1992. *NEFSC Ref. Doc.* 94-05. Woods Hole, MA.
Brodziak, J., and K. Sosebee. **1994.** Application of a biomass dynamics model to the spiny dogfish (*Squalus acanthias*). *NEFSC Ref. Doc.* 94-18. Woods Hole, MA.
Rago, P., K. Sosebee, **J. Brodziak**, and E. Anderson. **1994.** Distribution and dynamics of North Atlantic spiny dogfish (*Squalus acanthias*). *NEFSC Ref. Doc.* 94-19. Woods Hole, MA
Brodziak, J. 1994. A brief overview of the use of trawl survey data for stock assessment with emphasis on the ADAPT VPA tuning model. In T. Berger (ed), *Workshop on the Collection and use of trawl survey data for fisheries management*. Atlantic States Marine Fisheries Commission, Washington, D.C.
Brodziak, J.K.T., and P. J. Rago. **1994.** Application of stochastic dynamic programming to determine optimal harvesting strategies for the Georges Bank yellowtail flounder (*Pleuronectes ferrugineus*) stock. *A.F.S. 1994 Annual Meeting Abstracts*, Halifax, Nova Scotia, p. 158.
Hayes, D.B., and J.K.T. Brodziak. **1994.** Impact of discard mortality on sustainable fishing mortality. A.F.S. *1994 Annual Meeting Abstracts*, Halifax, Nova Scotia, p. 155.
Brodziak, J. 1993. An extension of stock composition analysis to include marking data. *Can. J. Fish. Aquat. Sci*. 50:251-257.
Brodziak, J.K.T., and A.A. Rosenberg. **1993.** A method to assess squid fisheries in the Northwest Atlantic. *ICES J. Mar. Sci.* 50:187-194.
Brodziak, J. 1993. Book Review: Modelling biological populations in space and time. Eric Renshaw, Cambridge Univ. Press. *Math. Biosci.* 114:249-250.
Overholtz, W. J., S. F. Edwards, and **J.K.T. Brodziak. 1993.** Strategies for rebuilding and harvesting New England groundfish resources. In G. Kruse *et al*. (eds) *Proceedings of the International Symposium on Management Strategies for Exploited Fish Populations*. Alaska Sea Grant College Program, AK-SG-93-02, Univ. of Alaska, Fairbanks.
Brodziak, J., B. Bentley, D. Bartley, G. A. E. Gall, R. Gomulkiewicz, and M. Mangel. **1992.** Tests of

genetic stock identification using coded wire tagged fish. *Can. J. Fish. Aquat. Sci.* 49:1507-1517.
Brodziak, J. 1992. Applying genetic stock identification to assess the ocean chinook fisheries south of Cape Falcon, Oregon. In T. Hassler (Ed.), *1990 Northeast Pacific chinook and coho salmon workshop proceedings*. Calif. Coop. Fish. Research Unit, Arcata, CA.
Gall, G.A.E., Bartley, D., B. Bentley, **J. Brodziak**, R. Gomulkiewicz, M. Mangel. **1992.** Geographic variation in population genetic structure of chinook salmon from California and Oregon. *Fish. Bull., U.S.* 90:77-100.
Brodziak, J. 1992. Stock assessment of the long-finned squid, *Loligo pealei*, in the Northwest Atlantic for 1991. *Papers of the Northeast Regional Stock Assessment Workshops*.Res. Doc. SAW 14/1, CRD-92-06. Woods Hole, MA.
Brodziak, J. 1992. Stock assessment of the short-finned squid, *Illex illecebrosus*, in the Northwest Atlantic for 1991. *Papers of the Northeast Regional Stock Assessment Workshops*. Res. Doc. SAW 14/2, CRD-92-07. Woods Hole, MA.
Brodziak, J., and A. Rosenberg. **1991.** Assessment of squid fisheries in the Northwest Atlantic. *International Council for the Exploration of the Sea. C.M. 1991*/K:9.
Brodziak, J. 1991. Stock assessment of Atlantic butterfish, *Peprilus triacanthus*, in the Northwest Atlantic. *Papers of the Northeast Regional Stock Assessment Workshops.* Res. Doc. SAW 12/4, CRD 91-03. Woods Hole, MA.
Brodziak, J. 1991. Stock assessment of the short-finned squid, *Illex illecebrosus*, in the Northwest Atlantic. *Papers of the Northeast Regional Stock Assessment Workshops*. Res. Doc. SAW 12/6, CRD-91-03. Woods Hole, MA.
Brodziak, J. 1991. Stock assessment of the long-finned squid, *Loligo pealei*, in the Northwest Atlantic. *Papers of the Northeast Regional Stock Assessment Workshops*. Res. Doc. SAW 12/7, CRD-91-03. Woods Hole, MA.
Brodziak, J. 1991. Bootstrap estimators of discard rates using domestic sea sampling data. *Papers of the Northeast Regional Stock Assessment Workshops.* Res. Doc. SAW 12/8, CRD-91-03. Woods Hole, MA.
Brodziak, J. 1990. Theoretical aspects of genetic stock identification. *Ph.D. dissertation, University of California*. Davis, California.
Gomulkiewicz, R., **J. Brodziak**, and M. Mangel. **1990.** Ranking loci for genetic stock identification using curvature methods. *Can. J. Fish. Aquat. Sci.* 47:611-619.
Gall, G. A. E., B. Bentley, C. Panattoni, E. Childs, C. Qi, S. Fox, M. Mangel, **J. Brodziak**, and R. Gomulkiewicz. **1989.** Chinook mixed fishery project 1986-1989. *California Department of Fish and Game*, Rancho Cordova, California 95670, 192 p.

REVIEWER

- **Associate Editor**: *Transactions of the American Fisheries Society*, 2007 – 2013.

- **International Editorial Board Member**: *South African Journal of Marine Science*, 1998 – 2003.

- **Invited Reviewer:** *Animal Biotelemetry, Aquatic Living Resources, Asian Fisheries Science Journal, The Biological Bulletin*, *Bulletin of Marine Science*, *Blue Ocean Institute, CalCOFI Reports*, *California &*

Maine & Maryland & Rhode Island Sea Grant College Programs, *Canadian Journal of Fisheries and Aquatic Sciences, Chaos Solitons and Fractals, Deep Sea Research II, Ecography, Ecological Applications, Ecological Modeling, Environmental Modelling & Software, Fish and Fisheries, Fisheries Research, Fishery Bulletin, Hudson River Foundation, ICES Journal of Marine Science, International Journal of Tropical Biology, Journal of Fish Biology, Journal of Northwest Atlantic Fisheries Science, Journal of Oceanology and Limnology, Journal of Sea Research, Marine Biology, Marine Ecological Progress Series, Marine Ecology, Marine and Coastal Fisheries, Marine and Freshwater Research, Marine Resource Economics, Mathematical Biosciences, Monterey Bay Aquarium, NMFS Stock Assessment Analytical Methods RFP, NMFS Fisheries and the Environment RFP, National Research Institute for Far Seas Fisheries, National Taiwan University, North American Journal of Fisheries Management, Oecologica, PLOS ONE, Saltonstall-Kennedy Grant Program, Scientific Reports, South African Journal of Marine Science, Transactions of the American Fisheries Society, United Kingdom Department for Environment Food and Rural Affairs, United Nations World Ocean Assessment, United States Global Climate Change Research Program, Zoological Studies*.

PROFESSIONAL SOCIETIES

American Fisheries Society,1990-present

American Institute of Research Fisheries Biologists,1998-2012

SPECIALIZED TRAINING

Facilitative Skills: An Essential Foundation for Collaborative Leadership. 2018. Pacific Center for Collaboration**,** Dr. D. Ching, University of Hawaii.

Emotional Intelligence Workshop. **2015.** American Management Association Continuing Education Course, New York State University.

How to Lead an Effective Technical Meeting. **2011.** American Fisheries Society Continuing Education Course**,** Dr. J. Berkson, University of Florida.

NOAA Leadership Training Program. **2009.** Workforce Management Office of NOAA.

Conflict Resolution Skills for Natural Resource Professionals. **2006.** American Fisheries Society Continuing Education Course**,** Dr. M. Fraidenberg, Dynamic Solutions Group LLC.

COMPETITIVE GRANTS

Title: ***MARINE DEBRIS AS INCIDENTAL BYCATCH IN THE HAWAII LONGLINE FISHERY***
Principal Investigators: Dr. Amy Uhrin, NOAA Marine Debris Program
Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Grantor: National Marine Sanctuaries, 2018-2019, $10,000

Title: ***SMALL-VESSEL SURVEY AND AUCTION SAMPLING TO ESTIMATE GROWTH AND MATURITY OF ETELINE SNAPPERS AND IMPROVE DATA-LIMITED STOCK ASSESSMENTS***
Principal Investigators: Robert Humphreys and Dr. Jon Brodziak,
Pacific Islands Fisheries Science Center
Grantor: NMFS S&T, 2014-2015, $73,520

Title: ***DOCUMENTATION AND TESTING OF AGEPRO***
Principal Investigators: Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Dr. Erik Franklin, University of Hawaii, Hawaii Institute of Marine Biology
Grantor: NMFS S&T, 2014-2015, $19,841

Title: ***DEVELOPMENT AND IMPLEMENTATION OF A SUITE OF LENGTH-BASED MORTALITY ESTIMATORS***
Principal Investigators: Dr. John Walter, Southeast Fisheries Science Center
Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Dr. John Hoenig, Virginia Institute of Marine Science
Grantor: NMFS S&T, 2014-2015, $75,303

Title: ***ANALYSES OF CATCH DATA FOR BLUE AND STRIPED MARLINS***
Principal Investigators: Dr. William Walsh, University of Hawaii, JIMAR
Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Grantor: UH Manoa, Pelagic Fisheries Research Program, $15,000

Title: ***QUANTIFYING THE INFORMATION CONTENT AND CONSISTENCY BETWEEN ALTERNATIVE SOURCES OF FISHERY DATA***
Principal Investigators: Prof. Marc Mangel, University of California, Santa Cruz
Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Grantor: NMFS PIFSC, 2012-2013, $40,000

Title: ***RESILIENCY OF STOCK-RECRUITMENT RELATIONSHIPS OF PACIFIC TUNAS TO ENVIRONMENTAL VARIATION***
Principal Investigators: Prof. Marc Mangel, University of California, Santa Cruz
Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Grantor: NMFS PIFSC, 2009-2011, $75,000

Title: ***RESILIENCY OF STOCK-RECRUITMENT RELATIONSHIPS OF PACIFIC TUNAS TO ENVIRONMENTAL VARIATION***
Principal Investigators: Prof. Marc Mangel, University of California, Santa Cruz
Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Grantor: NMFS PIFSC, 2009-2011, $75,000

Title: ***COMPARISONS OF TOTAL AND NATURAL MORTALITY RATES OF REEF SPECIES IN THE HAWAIIAN ARCHIPELAGO AND THE CARIBBEAN REGION.***
Principal Investigators: Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Dr. Clay Porch, Southeast Fisheries Science Center
Grantor: NMFS S&T, 2010-2011, $30,000

Title: ***USING SATELLITE DATA TO IMPROVE SHORT-TERM RECRUITMENT PREDICTIONS FOR MARINE FISH STOCKS.***
Principal Investigator: Dr. Jon Brodziak, Pacific Islands Fisheries Science Center
Grantor: NOAA NASA, 2007-2008, $15,000

Title: ***HADDOCK MIGRATION IN NEW ENGLAND WATERS: ANALYSIS OF MOVEMENTS BETWEEN STOCKS AND CLOSED AREAS.***
Principal Investigators: Paul Parker, Cape Cod Hook Fishermen's Association
Dr. Jon Brodziak, Northeast Fisheries Science Center
Dr. Shelly Tallack, Gulf of Maine Research Institute
Grantor: Northeast Consortium, 2005-2006, $210,000

Title: ***CLIMATE-BASED ASSESSMENT AND FORECASTS FOR ECOSYSTEMS IN THE GULF OF MAINE.***
Principal Investigators: Drs. Andrew Pershing and Charles Greene, Cornell University
Associate Investigators: Drs. Barbara Bailey, Jon Brodziak and J. Jossi.
Grantor: NOAA Fisheries Coastal Ocean Program, 2003-2004, $140,000

Title: ***RESOLVING POPULATION STRUCTURE WITH MOLECULAR GENETICS TO ENHANCE MANAGEMENT OF THE HIGHLY EXPLOITED SQUID FISHERY.***
Principal Investigator: Dr. Roger Hanlon, Marine Biological Laboratory
Associate Investigators: Dr. Jon Brodziak, Kendra Buresch, Steve Cadrin, Arnold Howe
Grantor: Woods Hole Oceanographic Institution Sea Grant, 2002-2003, $50,000

Title: ***TROPHIC DYNAMICS OF THE LONG-FINNED SQUID ON GEORGES BANK: FEEDING AND GROWTH IN RELATION TO MULTISPECIES FISHERIES MANAGEMENT.***
Principal Investigators: Drs. W. K. Macy, III and A. Durbin, University of Rhode Island
Associate Investigator: Dr. Jon Brodziak
Grantor: Rhode Island Sea Grant, 1993-1995, $67,000

REFERENCES

Professor Marc Mangel, Ph.D.
Distinguished Professor of Mathematical Biology, Emeritus, and Former Director of the Center for Stock Assessment Research in Department of Applied Mathematics and Statistics, University of California, Santa Cruz, Santa Cruz, CA 95064
(408) 459-5797
Email: msmangel@ams.ucsc.edu

Professor Steven Murawski, Ph.D.
Professor and Peter Betzer Endowed Chair in Biological Oceanography
College of Marine Science, University of South Florida, FL 33701
(727) 553-3367
Email: smurawski@usf.edu

Professor Chi-Lu Sun, Ph.D.
Professor Emeritus, Center of Excellence for the Oceans
National Taiwan Ocean University, 2 Beining Road
Keelung 20224, Taiwan
Email: chilu@ntu.edu.tw